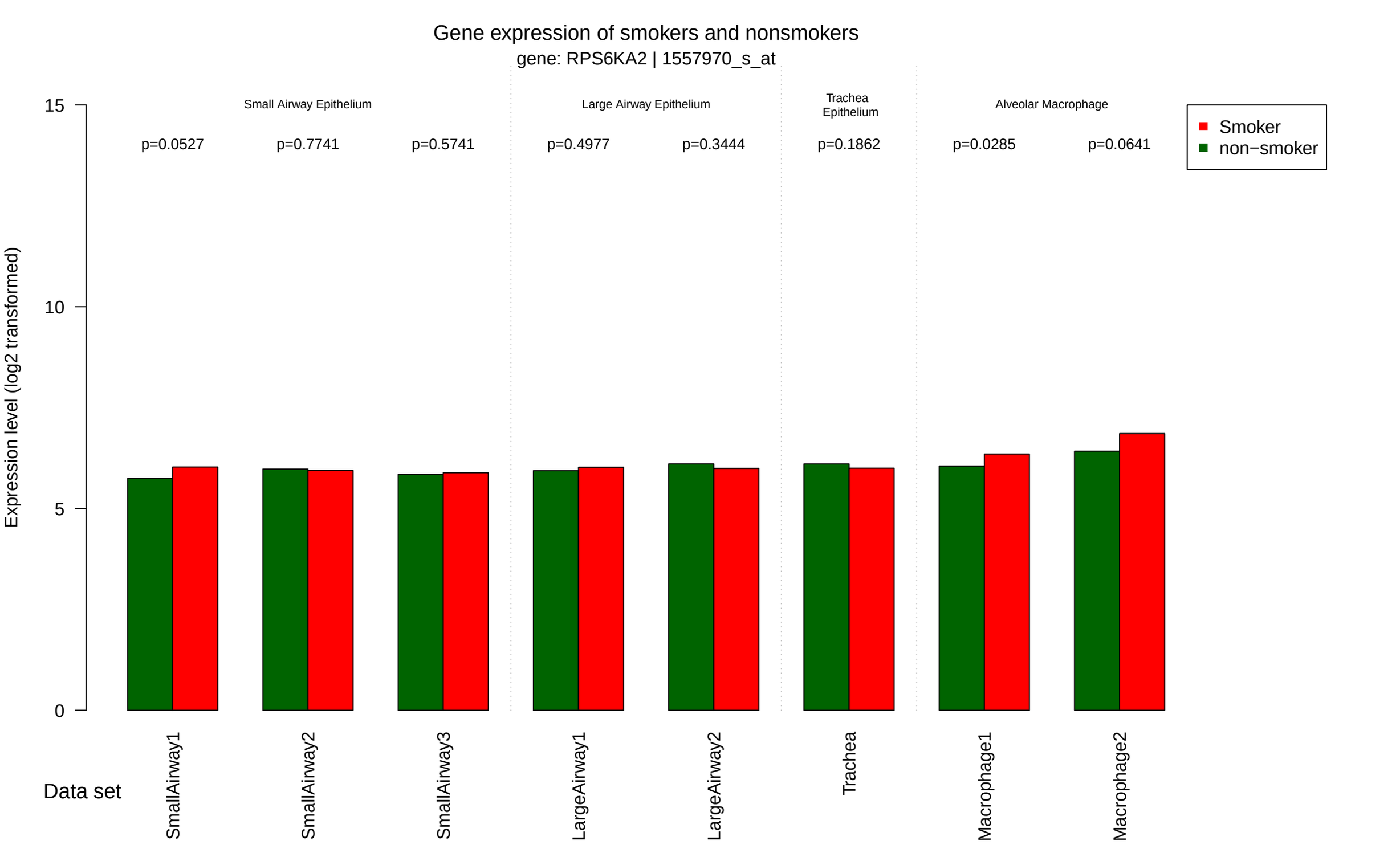
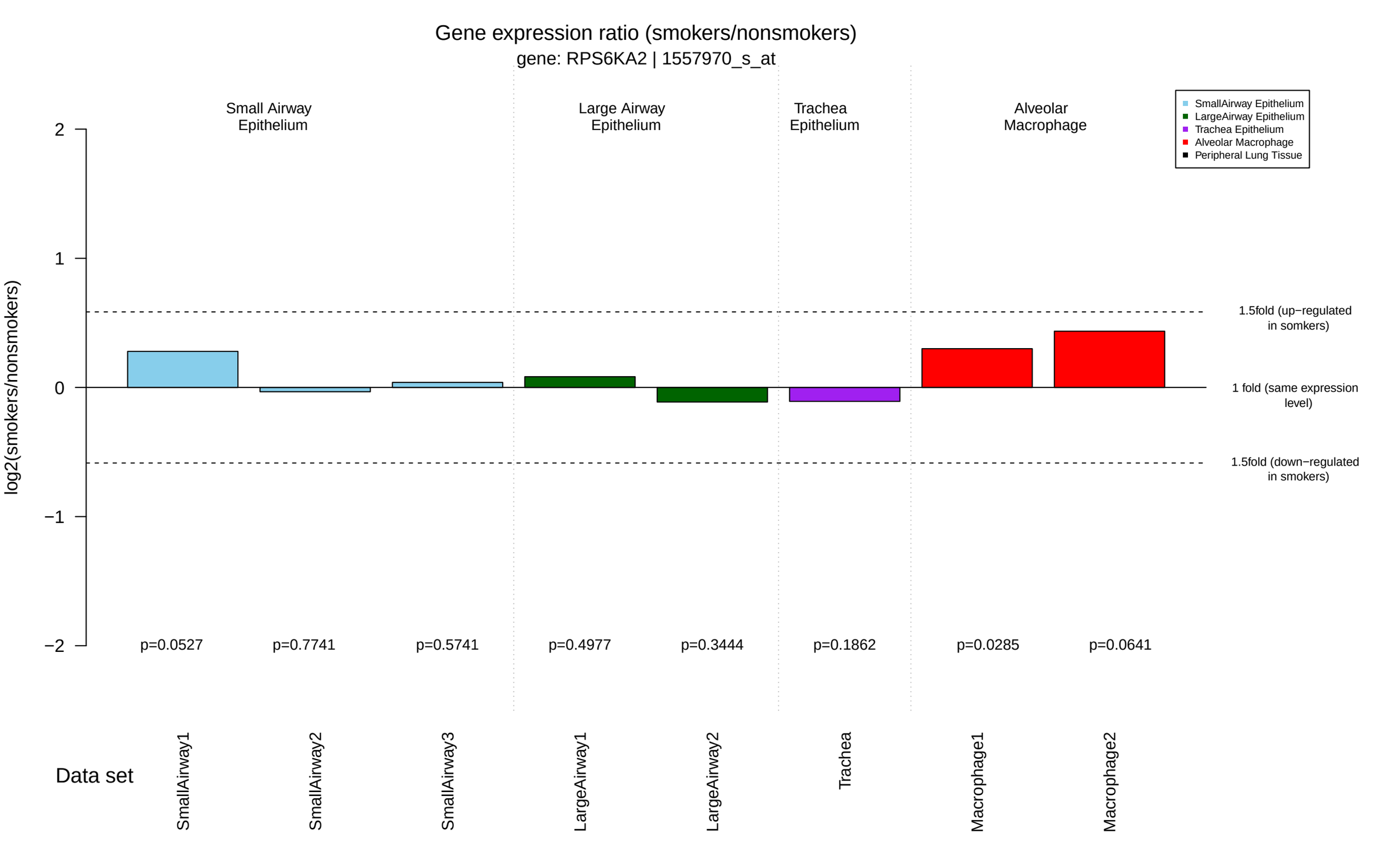
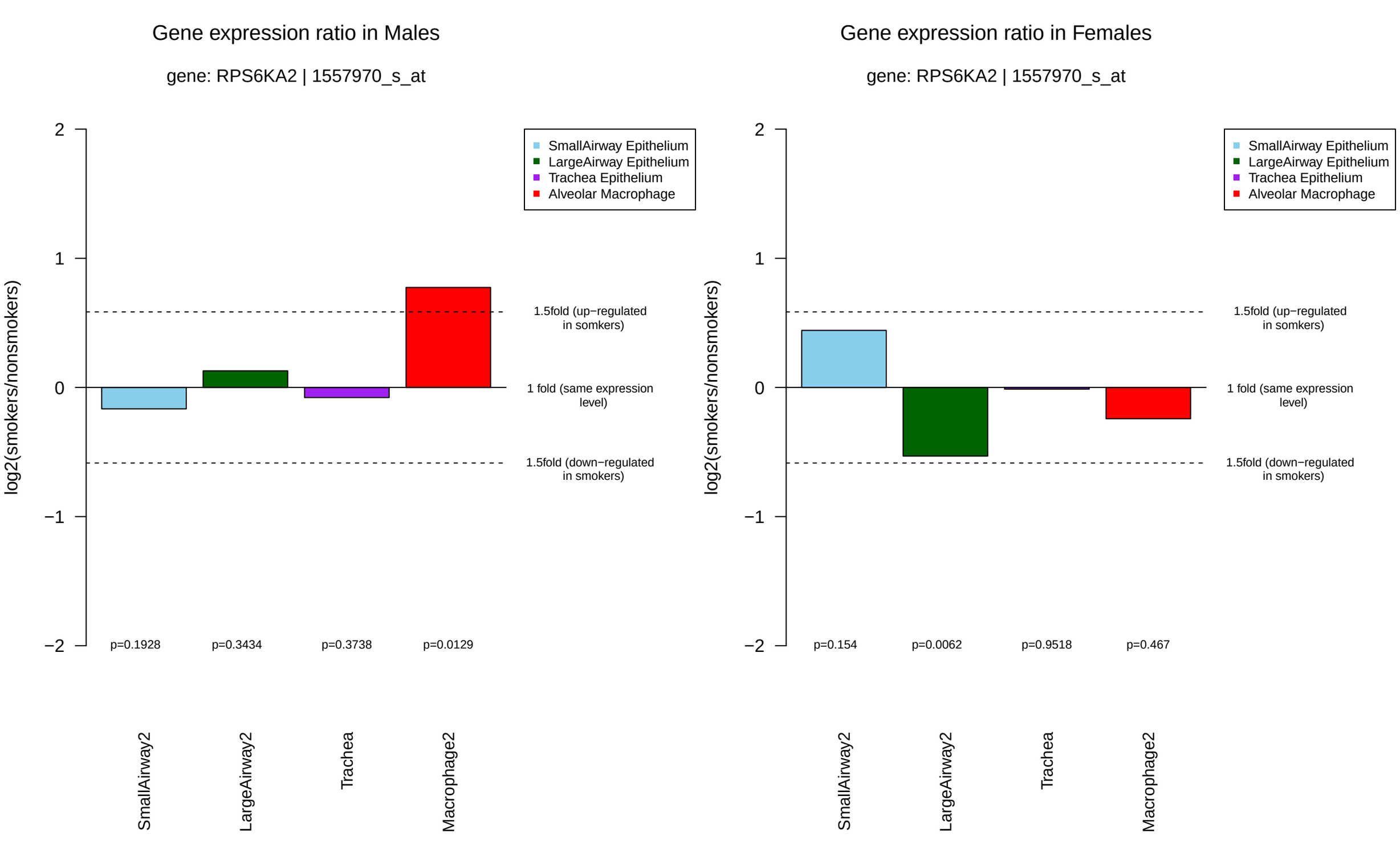
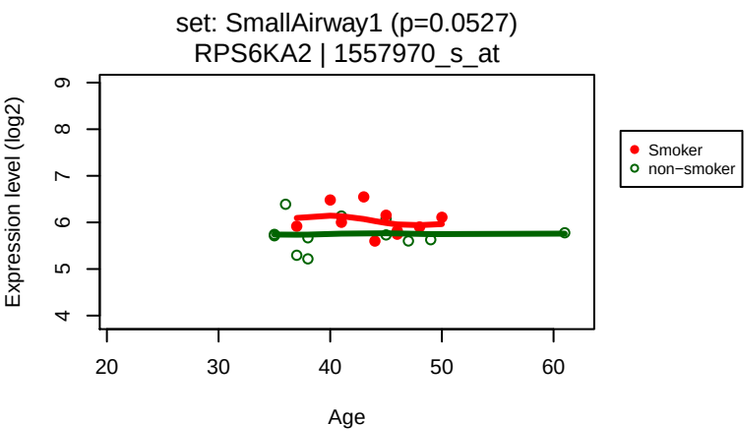
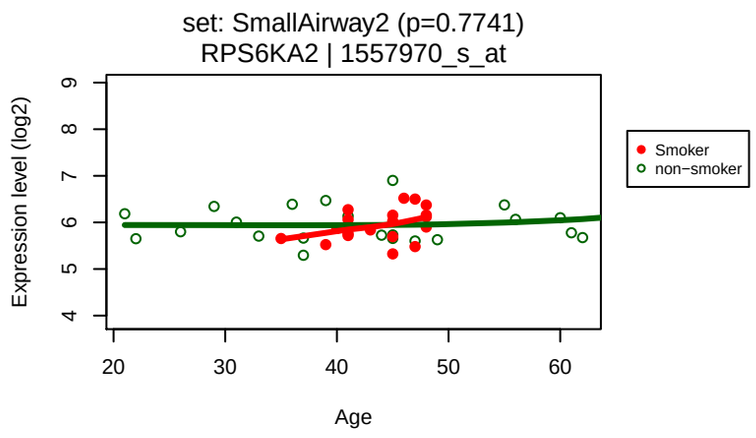
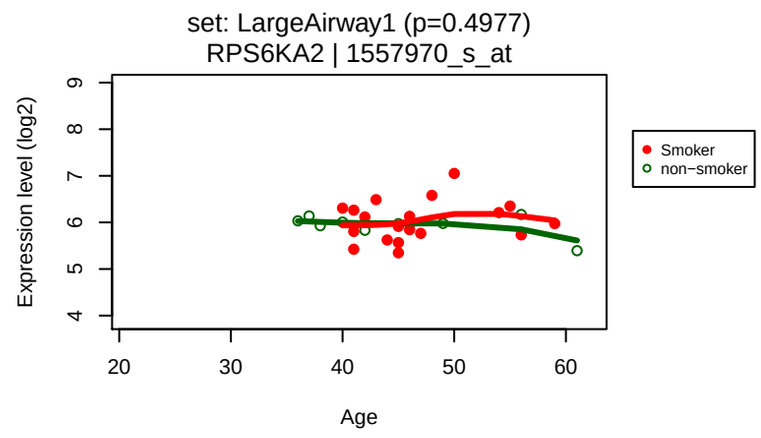

Gene expression of smokers and nonsmokers
gene: RPS6KA2 | 1557970_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0527
p=0.7741
p=0.5741
p=0.4977
p=0.3444
p=0.1862
p=0.0285
p=0.0641
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

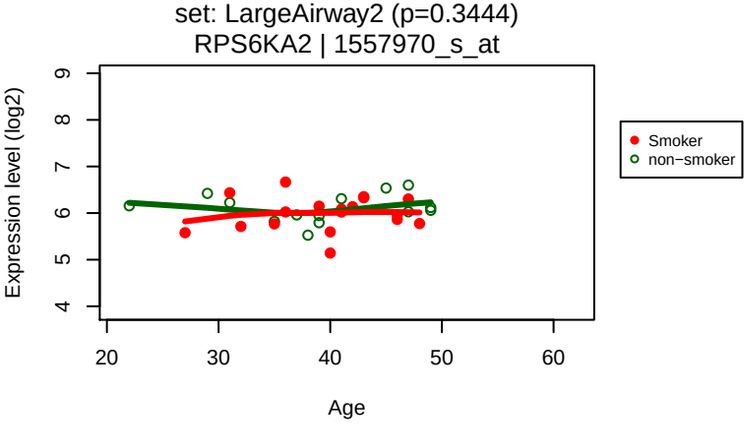

Gene expression ratio (smokers/nonsmokers)
gene: RPS6KA2 | 1557970_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0527
p=0.7741
p=0.5741
p=0.4977
p=0.3444
p=0.1862
p=0.0285
p=0.0641
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

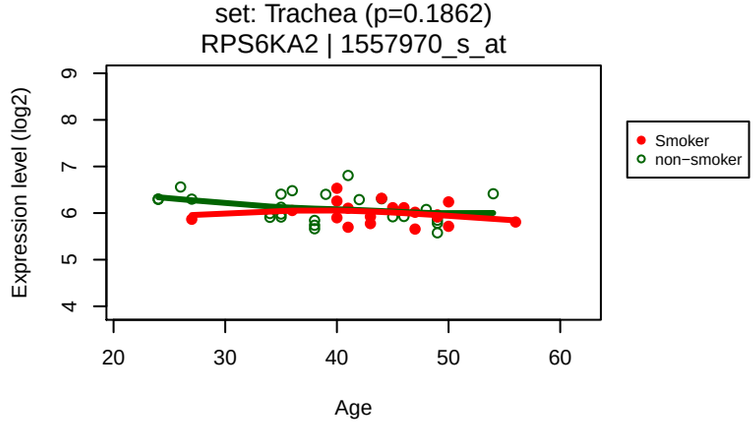

Gene expression ratio in Males
gene: RPS6KA2 | 1557970_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1928
p=0.3434
p=0.3738
p=0.0129
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPS6KA2 | 1557970_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.154
p=0.0062
p=0.9518
p=0.467
SmallAirway2
LargeAirway2
Trachea
Macrophage2

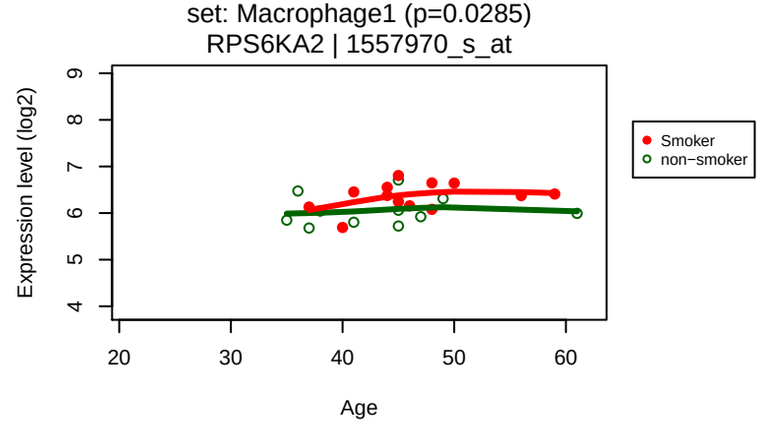

set: SmallAirway1 (p=0.0527)
RPS6KA2 | 1557970_s_at
Expression level (log2)
Age
Smoker
non−smoker

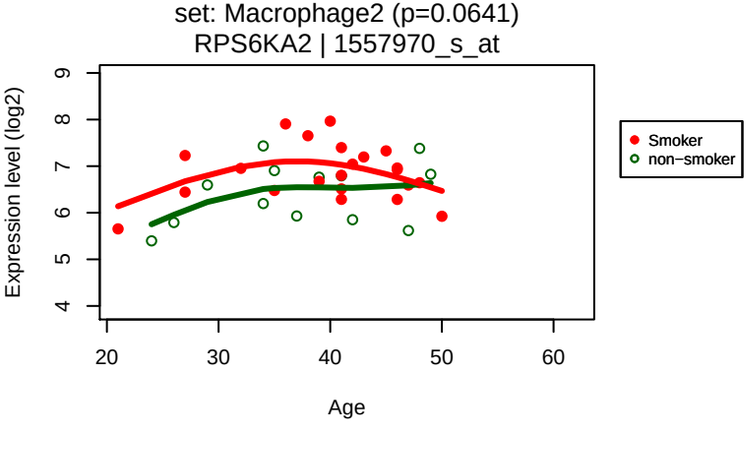

set: SmallAirway2 (p=0.7741)
RPS6KA2 | 1557970_s_at
Expression level (log2)
Age
Smoker
non−smoker

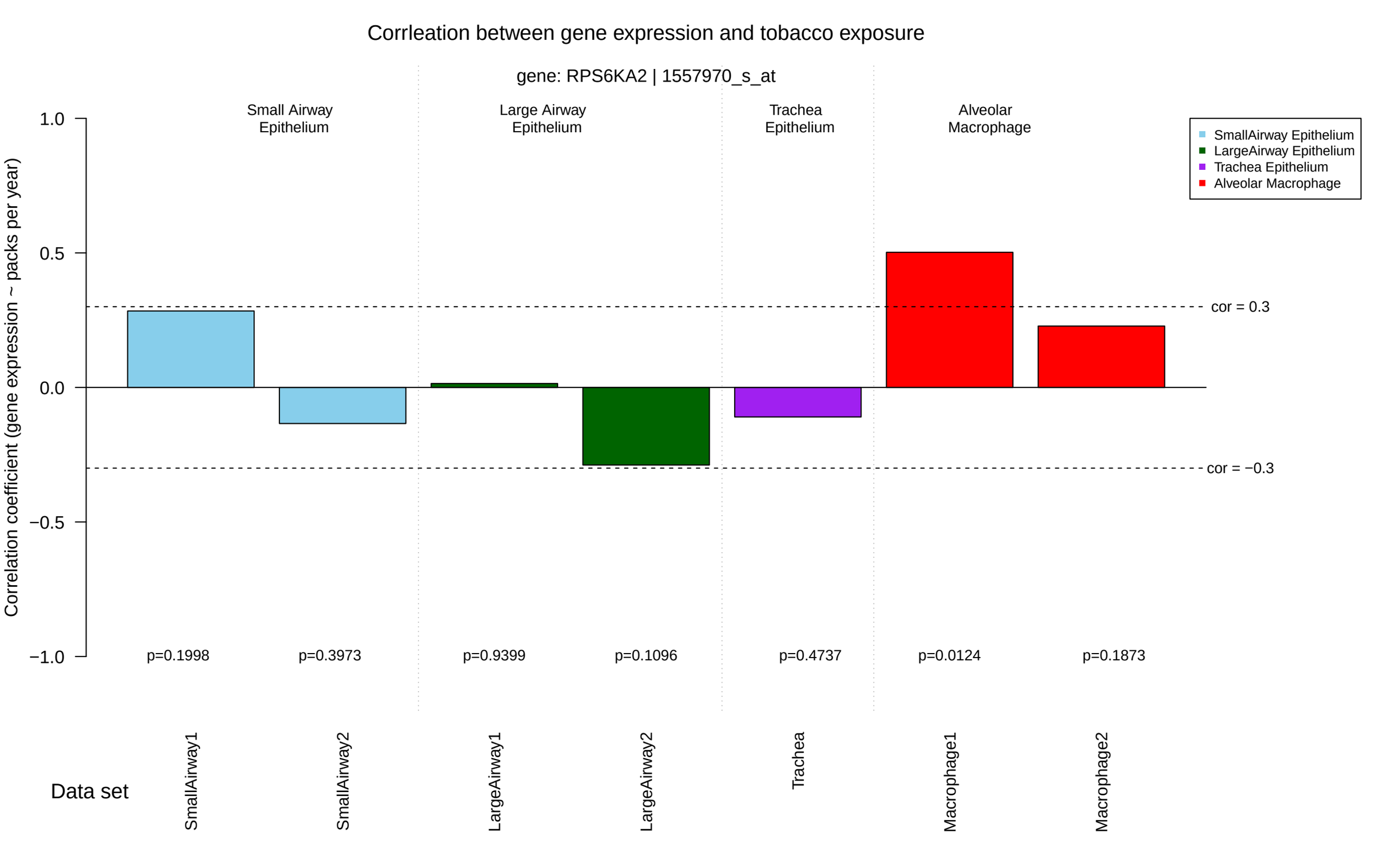

set: LargeAirway1 (p=0.4977)
RPS6KA2 | 1557970_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3444)
RPS6KA2 | 1557970_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1862)
RPS6KA2 | 1557970_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0285)
RPS6KA2 | 1557970_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0641)
RPS6KA2 | 1557970_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPS6KA2 | 1557970_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1998
p=0.3973
p=0.9399
p=0.1096
p=0.4737
p=0.0124
p=0.1873
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2